



VELS





INSTITUTE OF SCIENCE, TECHNOLOGY & ADVANCED STUDIES (VELS)
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MOODTUNE AI: SONG EMOTION ANALYSIS AND EMOTION-BASED MUSIC RECOMMENDATION USING DISTILBERT

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- Most music recommendation systems rely on listening history or metadata —
not the lyrics or emotional content of a song.
- This approach fails for new or unpopular tracks that have no listening history yet.
- Lyrics carry rich emotional signals that NLP models can extract directly from the text.
- MoodTune AI reads a song’s lyrics, classifies its emotion using a DistilBERT-based model, and recommends similar songs based on that emotional profile.

Abstract



01 Takes a song name as input, retrieves its lyrics, and classifies the dominant emotion into 6 categories: Happy, Sad, Love, Anger, Fear, Motivational.

02 Uses a DistilBERT model fine-tuned on 5,910 lyrics songs, with class weighting loss and dataset expansion to handle severely imbalanced emotion categories.

03 Displays a confidence score (softmax probability) with every prediction.

04 Recommends 5 other songs sharing the same predicted emotion, each linked to YouTube for playback.

05 Final model achieves 67.6% accuracy , with Motivational F1-score improved from 0.00 to 0.52- through iterative fixes: class-weighted loss training, and manually expanding the Motivational category from 44 to 75 songs.

BASE PAPER

Paper	LyEmoBERT-Revathy V.R., Anitha S, Pillai, Fatemah Daneshfar (2023)
Contribution	Fine-tuned BERT-Base model to classify song lyrics into 4 emotions and recommend similar songs using a Sentence Transformer.
Methodology	Transfer learning: trained a simple classifier (Naïve Bayes) on a small human-labeled dataset (MER, 770 songs), used it to autolabel a larger dataset (Music4All), then fine-tuned BERT on the newly labeled data.
Dataset	Music4All (28,000 songs, later filtered to ~28K lyrics), 4 emotions: Happy, Sad, Angry, Relaxed.
Limitations	Only 4 emotions (no Love/Motivational); full BERT is computationally heavy; no confidence score shown to users; class imbalance not explicitly handled (Relaxed class had lowest accuracy, ~69%).

LITERATURE SURVEY

Author & Title	Methodology	Limitations	Our Contribution
Agrawal, Shanker & Alluri (2021)	Transformer-based text encoder for emotion recognition from lyrics	Focuses on classification accuracy without addressing severe class imbalance in emotion categories	We specifically tackle a severely underrepresented emotion category (Motivational) through class-weighted training and dataset expansion
Song & Beck (2022)	Combines BERT with emotion lexicons and a State Space Model to track emotion changes verse-by-verse within a song	Models fine-grained emotional intensity over time rather than a single overall label, and performs poorly on rare emotions like Surprise	We classify a song's single overall dominant emotion, which is what a recommendation system needs, and directly address our own rare-category problem (Motivational)
Sakunkoo & Sakunkoo (2024)	**Trains a CNN on out-of-domain Reddit comments, due to a lack of labeled song lyrics data, then applies it to classify song emotions**	Relies on out-of-domain training data as a workaround for unlabeled lyrics, and is evaluated on only 100 songs	We apply an emotion classifier directly to our own 5,910-song dataset, evaluated at a much larger scale
Pizarro Martinez et al (2024)	DistilBERT applied to genre and commercial-success prediction	DistilBERT not previously applied to emotion-based recommendation	First application of DistilBERT to lyric-based emotion classification integrated with a functional recommendation engine.
Cui (2025)	Applies a weighted loss function to BERT, RoBERTa, and BART to address class imbalance in multi-label emotion detection	Found that weighted loss mainly helps high-frequency classes, with limited improvement on minority emotions	**Our own results show the same pattern — weighted loss alone raised our rare-category F1 from 0.00 to only 0.42, which is why we also added real training examples rather than relying on weighting alone**

MoodTune AI — System Architecture

Layered architecture (client, application, ML model, data) with the offline training subsystem alongside

CLIENT LAYER

Web Browser
User enters song title, views results

HTML / CSS / JavaScript Front-End
Renders emotion, confidence, recommendations

APPLICATION LAYER (Flask)

song_title

JSON response

Request Handler
/analyze route, validates song_title

Inference Orchestrator
Coordinates lookup, predict, recommend

Response Builder
Formats emotion, confidence, songs

Flask Route Layer
/analyze endpoint (HTTP POST)

ML MODEL LAYER

Tokenizer
DistilBERT WordPiece, max_length 256

Fine-tuned DistilBERT
moodtune_model_final, 6-class head

Softmax + Argmax
Probability to label and confidence

Recommendation Engine
Filters by emotion, samples 5 songs

DATA LAYER

songs_labeled_v2.csv
5,910 songs: lyrics and emotion label

moodtune_model_final/
Deployed model weights (produced by training, right)

OFFLINE TRAINING SUBSYSTEM

Raw Dataset
6,027 songs, 21 artists

Preprocess + Auto-Label
Clean, truncate to 6,000; classify to 5,910-song pool

Class Imbalance Fix
Weighted loss (49x) plus 31 new songs

4 Training Experiments
Baseline, Resample, Weighted, Expansion

Evaluation
Accuracy, F1, confusion matrix

produces

MoodTune AI — System Modules

Eight modules, from raw data collection to live song recommendation

OFFLINE — DATA & TRAINING

1

Data Collection

Kaggle dataset, 21 artists, 6,027 songs combined

2

Data Preprocessing

Cleans titles, truncates to 6,000, removes 71 explicit songs (5,910 usable)

3

Emotion Labeling (Automated)

No human labels available; pretrained classifier, multi-chunk sampling + majority voting

4

Class Imbalance Handling

Custom WeightedTrainer, inverse-frequency weights (Motivational 49x Happy)

LIVE — APPLICATION

5

Model Training

Fine-tunes DistilBERT across 4 experiments (Baseline, Resample, Weighted, Expansion), CPU-only

6

Model Evaluation

Accuracy + per-class F1-scores, confusion matrix analysis

7

Flask Web Application

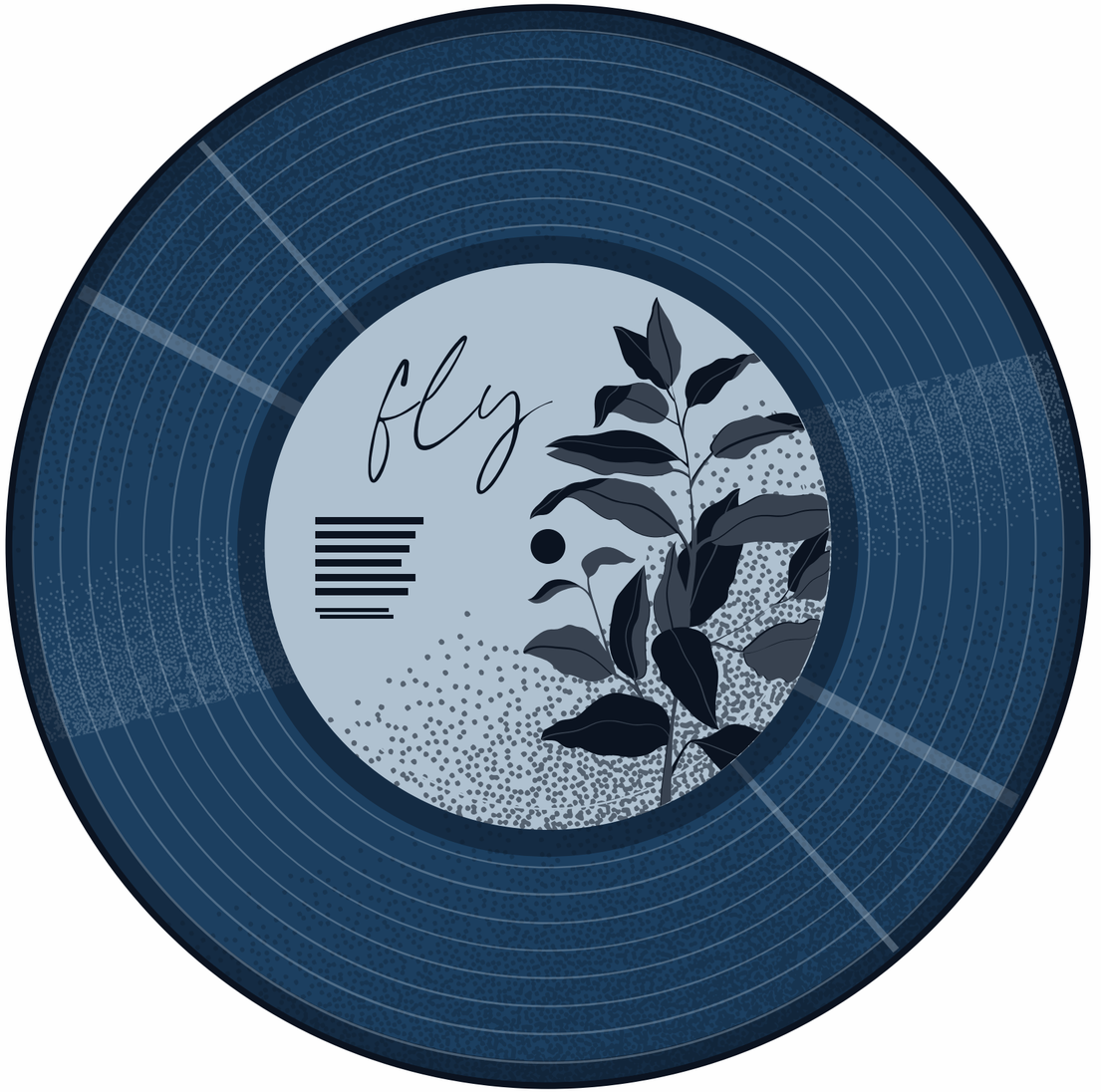
Accepts song title, retrieves lyrics, runs inference, returns prediction + confidence

8

Recommendation

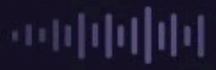
Samples 5 same-emotion songs from the pool, links each to YouTube

Six Emotions. One Model. Zero Cherry-Picking.



MoodTune AI

Hear what your lyrics feel.



Try thank u, next or Dangerous Woman

Analyze

Stronger

▶ Play

MOTIVATIONAL

95.47%

RECOMMENDED - SAME MOOD

01	Someone Else	▶ Play
02	Stronger (What Doesn't Kill You)	▶ Play
03	Teardrops On My Guitar (Acoustic)	▶ Play
04	Eye of the Tiger	▶ Play
05	We Are the Champions	▶ Play

MoodTune AI

Hear what your lyrics feel.



Try thank u, next or Dangerous Woman

Analyze

Heartbeat

▸ Play

LOVE

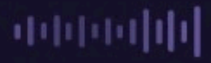
97.12%

RECOMMENDED - SAME MOOD

01	Single Ladies (Remix)	▸ Play
02	Blueberry Kisses	▸ Play
03	Cameras	▸ Play
04	Love on the Brain	▸ Play
05	Lounge Act	▸ Play

MoodTune AI

Hear what your lyrics feel.



Try thank u, next or Dangerous Woman

Analyze

Firework

▸ Play

HAPPY

94.99%

RECOMMENDED - SAME MOOD

01	Trophies	▸ Play
02	limbo*	▸ Play
03	Three Little Birds	▸ Play
04	Work It	▸ Play
05	Miss Me (Like That)	▸ Play

MoodTune AI

Hear what your lyrics feel.



Try thank u, next or Dangerous Woman

Analyze

Dangerous Woman

▸ Play

ANGER

95.58%

RECOMMENDED - SAME MOOD

01	Real Pimpin	▸ Play
02	Could Never Be Mad	▸ Play
03	Hymn for the Weekend (Live in Buenos Aires)	▸ Play
04	Hoes Mad	▸ Play
05	Bad Meets Evil (Original)	▸ Play

MoodTune AI

Hear what your lyrics feel.



Try thank u, next or Dangerous Woman

Analyze

bury a friend

▸ Play

FEAR

98.45%

RECOMMENDED - SAME MOOD

01	Speechless	▸ Play
02	OUTRO: The Journey	▸ Play
03	A L I E N S	▸ Play
04	Paradise - radio edit	▸ Play
05	Justin Bieber - Mistletoe (Christian Lalama REMIX)	▸ Play

MoodTune AI

Hear what your lyrics feel.



Try thank u, next or Dangerous Woman

Analyze

Fix You

▸ Play

SAD

97.88%

RECOMMENDED - SAME MOOD

01	Work (Bad Royale Remix) [Clean]	▸ Play
02	8TEEN	▸ Play
03	Make It Rain	▸ Play
04	Change	▸ Play
05	Unforgettable	▸ Play

RESULTS SUMMARY

We ran four training experiments. The baseline couldn't learn the Motivational category at all — its F1-score was 0.00, meaning it never once got a motivational song right. We tried resampling next, which actually made things worse — accuracy dropped to 51%. That told us the real problem was a shortage of examples, not just imbalance. Weighting the loss function was our first real breakthrough — Motivational's F1 rose to 0.42. Finally, we manually added 31 more genuine motivational songs, pushing F1 up to 0.52 — our best result.

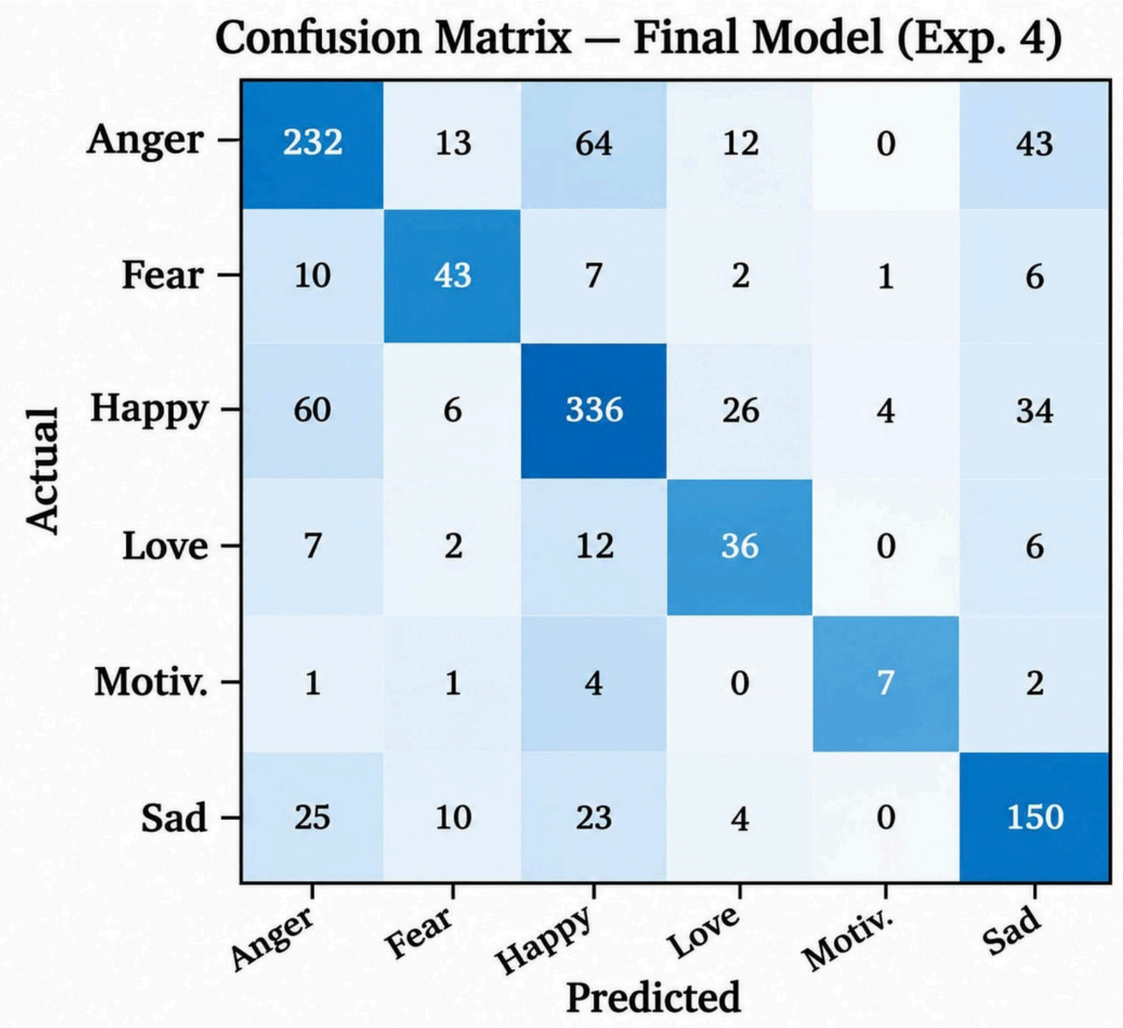
Summary Comparison of All Four Training Experiments

Experiment	Accuracy	Motiv. F1
1. Baseline (Unweighted)	72%	0.00
2. Resampling (Rejected)	51%	0.00
3. Class-Weighted Loss	70%	0.42
4. Dataset Expansion (Final)	**67.6%**	**0.52**

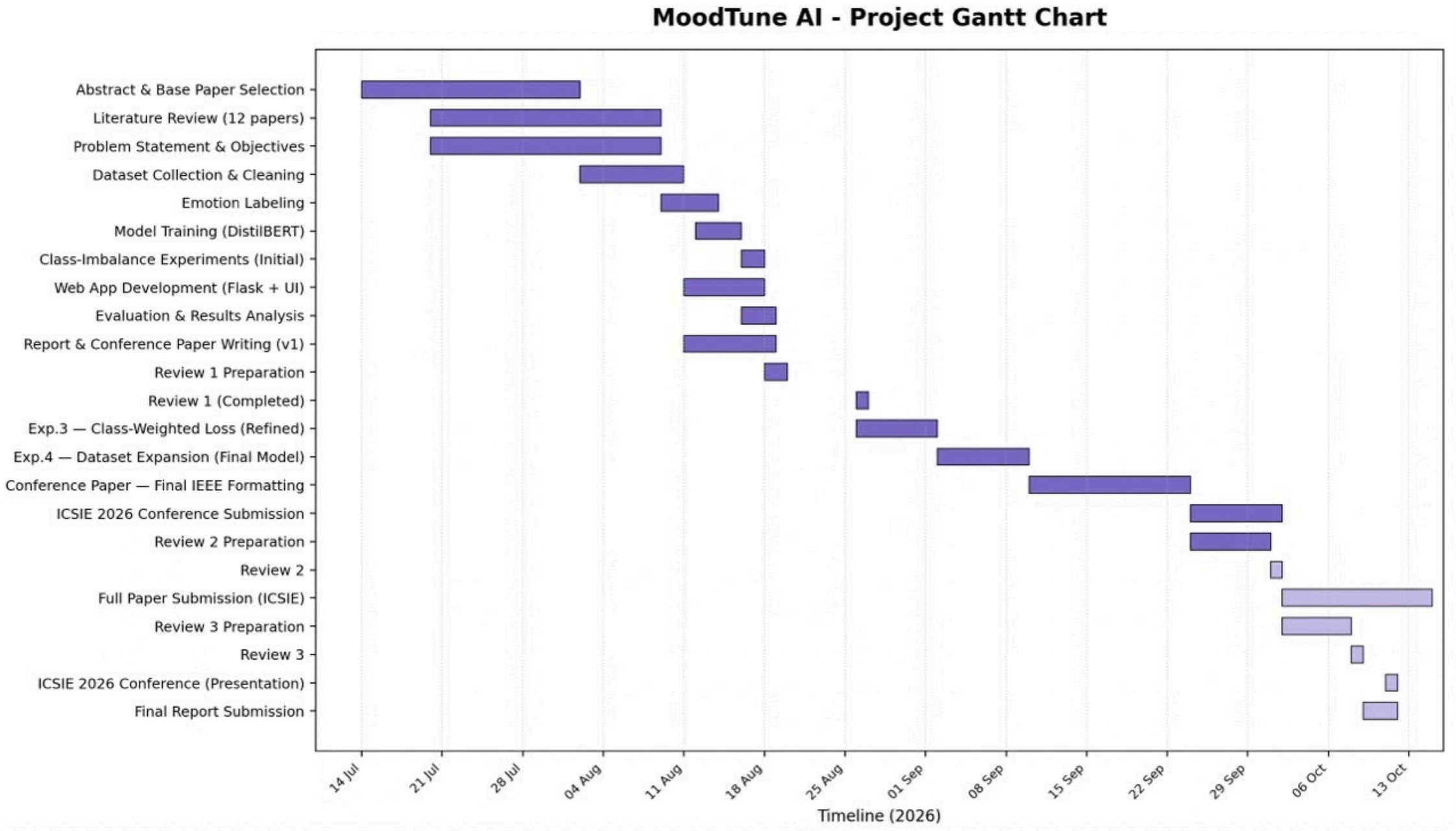
WHY DID ACCURACY DROP

- Our third experiment reached 70% accuracy using weighted loss, but Motivational's F1-score was only 0.42 — the model still struggled to catch enough real motivational songs. In our fourth experiment, we manually added 31 more genuine motivational songs, which pushed Motivational's F1-score up to 0.52 — a real improvement in the category we were specifically trying to fix.

- The overall accuracy dropped slightly, from 70% to 67.6%. But when we looked at the confusion matrix, that drop wasn't caused by our fix — it came from an unrelated, pre-existing issue: the model sometimes confuses Anger with Happy or Sad. That problem existed before we even touched the Motivational category, so it's a separate limitation we've noted for future work, not a side-effect of what we changed.



Project Timeline



LIMITATIONS

Two limitations are worth being upfront about. First, our labels are automatically generated, not human-verified — so some noise in the data is expected. Second, the emotion classifier we built on top of wasn't designed for music at all; it was trained on general text. We actually saw this ourselves today — a song literally called 'Love' came back as Happy, and one of our Fear-labeled songs used romantic language like 'trembling' that the model picked up as fear-adjacent instead.



Future Work

Grow the Motivational dataset further, investigate the Anger/Happy/Sad confusion (biggest remaining opportunity), explore multilingual lyrics and audio-feature integration.



- MoodTune AI demonstrates that a lightweight model like DistilBERT can meaningfully classify song emotions from lyrics alone, even starting from a severely imbalanced, automatically-labeled datas

- The final deployed model achieves 67.6% accuracy across all six emotions, verified working live in the Flask app.

THANK YOU